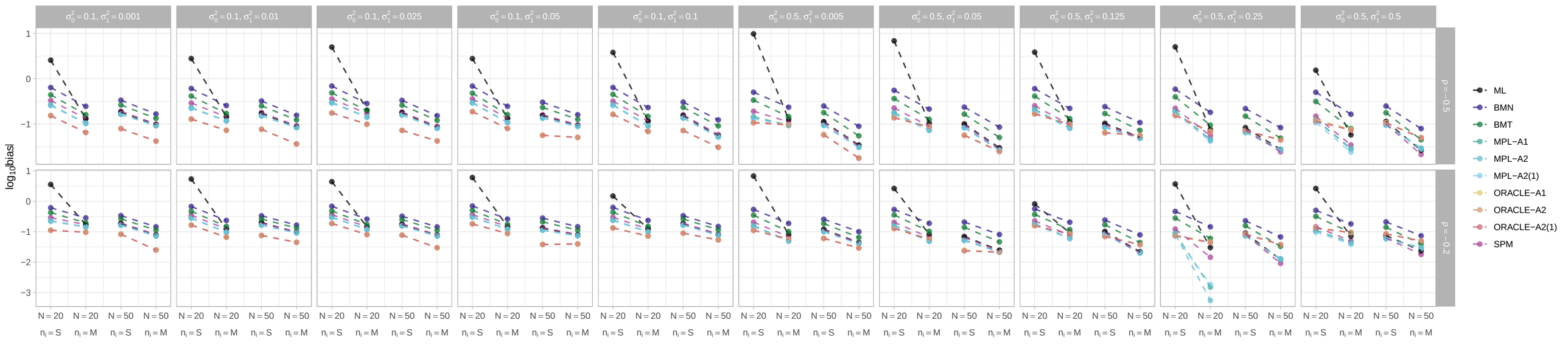
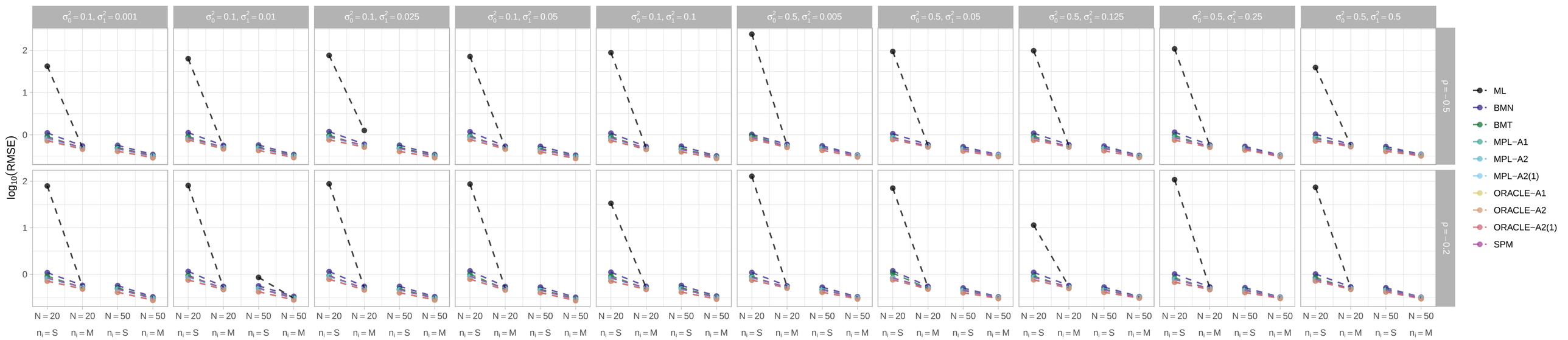
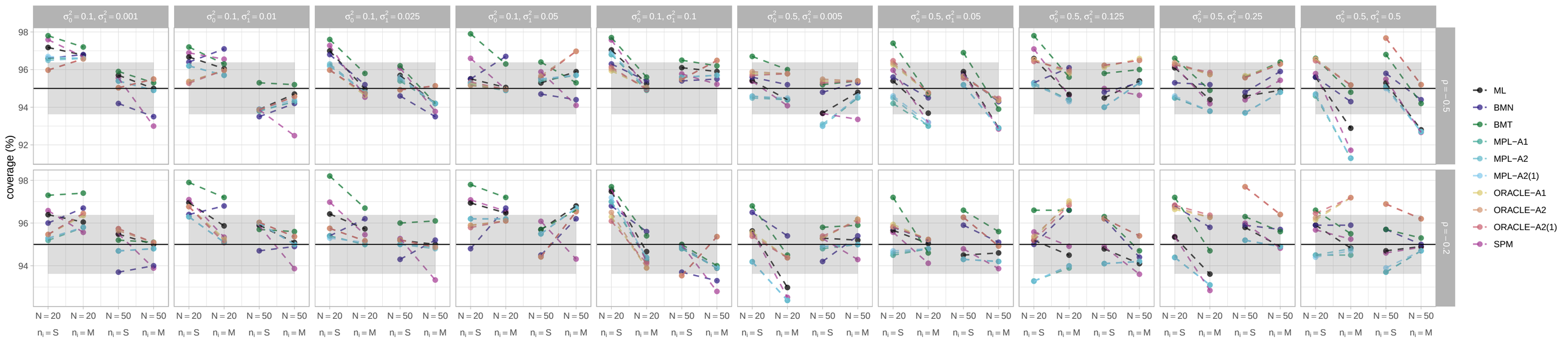
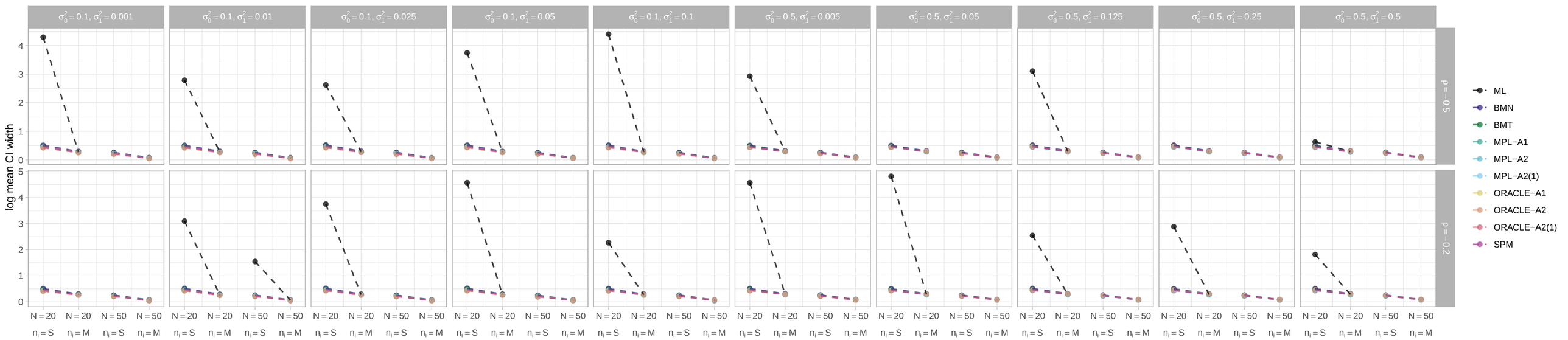

$\sigma_0^2 = 0.1, \sigma_1^2 = 0.001$
$\sigma_0^2 = 0.1, \sigma_1^2 = 0.01$
$\sigma_0^2 = 0.1, \sigma_1^2 = 0.025$
$\sigma_0^2 = 0.1, \sigma_1^2 = 0.05$
$\sigma_0^2 = 0.1, \sigma_1^2 = 0.1$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.005$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.05$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.125$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.25$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.5$
$\rho = -0.5$
$\rho = -0.2$
$\log_{10}$|bias|
1
0
−1
1
0
−1
−2
−3
N = 20 N = 20 N = 50 N = 50
n_i = S n_i = M n_i = S n_i = M
ML
BMN
BMT
MPL−A1
MPL−A2
MPL−A2(1)
ORACLE−A1
ORACLE−A2
ORACLE−A2(1)
SPM


$\sigma_0^2 = 0.1, \sigma_1^2 = 0.001$
$\sigma_0^2 = 0.1, \sigma_1^2 = 0.01$
$\sigma_0^2 = 0.1, \sigma_1^2 = 0.025$
$\sigma_0^2 = 0.1, \sigma_1^2 = 0.05$
$\sigma_0^2 = 0.1, \sigma_1^2 = 0.1$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.005$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.05$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.125$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.25$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.5$
$\rho = -0.5$
$\rho = -0.2$
$\log_{10}$(RMSE)
2
1
0
2
1
0
N = 20 N = 20 N = 50 N = 50
n_i = S n_i = M n_i = S n_i = M
ML
BMN
BMT
MPL−A1
MPL−A2
MPL−A2(1)
ORACLE−A1
ORACLE−A2
ORACLE−A2(1)
SPM


$\sigma_0^2 = 0.1, \sigma_1^2 = 0.001$
$\sigma_0^2 = 0.1, \sigma_1^2 = 0.01$
$\sigma_0^2 = 0.1, \sigma_1^2 = 0.025$
$\sigma_0^2 = 0.1, \sigma_1^2 = 0.05$
$\sigma_0^2 = 0.1, \sigma_1^2 = 0.1$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.005$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.05$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.125$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.25$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.5$
$\rho = -0.5$
$\rho = -0.2$
coverage (%)
98
96
94
98
96
94
N = 20 N = 20 N = 50 N = 50
n_i = S n_i = M n_i = S n_i = M
ML
BMN
BMT
MPL−A1
MPL−A2
MPL−A2(1)
ORACLE−A1
ORACLE−A2
ORACLE−A2(1)
SPM


$\sigma_0^2 = 0.1, \sigma_1^2 = 0.001$
$\sigma_0^2 = 0.1, \sigma_1^2 = 0.01$
$\sigma_0^2 = 0.1, \sigma_1^2 = 0.025$
$\sigma_0^2 = 0.1, \sigma_1^2 = 0.05$
$\sigma_0^2 = 0.1, \sigma_1^2 = 0.1$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.005$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.05$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.125$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.25$
$\sigma_0^2 = 0.5, \sigma_1^2 = 0.5$
$\rho = -0.5$
$\rho = -0.2$
log mean CI width
4
3
2
1
0
5
4
3
2
1
0
N = 20 N = 20 N = 50 N = 50
n_i = S n_i = M n_i = S n_i = M
ML
BMN
BMT
MPL−A1
MPL−A2
MPL−A2(1)
ORACLE−A1
ORACLE−A2
ORACLE−A2(1)
SPM